

Problem Set 12
Due: May. 24, 2021

1. Section 5.1 Problem 1.

A statistician wants to estimate the mean height h (in meters) of a population, based on n independent samples $X_1, \dots, X_n$, chosen uniformly from the entire population. He uses the sample mean $M_n = (X_1 + \dots + X_n)/n$ as the estimate of h , and a rough guess of 1.0 meters for the standard deviation of the samples X_i .

- (a). How large should n be so that the standard deviation of M_n is at most 1 centimeter?
- (b). How large should n be so that Chebyshev's inequality guarantees that the estimate is within 5 centimeters from h , with probability at least 0.99?
- (c). The statistician realizes that all persons in the population have heights between 1.4 and 2.0 meters, and revises the standard deviation figure that he uses based on the bound of Example 5.3. How should the values of n obtained in parts (a) and (b) be revised?

Solution:

- (a). We have $\sigma_{M_n} = 1/\sqrt{n}$, so in order that $\sigma_{M_n} \leq 0.01$, we must have $n \geq 10000$.
- (b). We want to have

$$\mathbf{P}(|M_n - h| \leq 0.05) \geq 0.99$$

Using the facts $h = \mathbf{E}[M_n]$, $\sigma_{M_n}^2 = 1/n$, and the Chebyshev inequality, we have

$$\begin{aligned}\mathbf{P}(|M_n - h| \leq 0.05) &= \mathbf{P}(|M_n - \mathbf{E}[M_n]| \leq 0.05) \\ &= 1 - \mathbf{P}(|M_n - \mathbf{E}[M_n]| \geq 0.05) \\ &\geq 1 - \frac{1/n}{0.05^2}\end{aligned}$$

Thus, we must have

$$1 - \frac{1/n}{0.05^2} \geq 0.99$$

which yields $n \geq 40000$.

- (c). Based on Example 5.3, $\sigma_{X_i}^2 \leq 0.6^2/4 = 0.09$, so he should use 0.3 meters in place of 1.0 meters as the estimate of the standard deviation of the samples X_i in the calculations of parts (a) and (b).

In the case of part (a), we have $\sigma_{M_n} = 0.3/\sqrt{n}$, so in order that $\sigma_{M_n} \leq 0.01$, we must have $n \geq 900$.

In the case of part (b), we have

$$1 - \frac{0.09/n}{0.05^2} \geq 0.99$$

which yields $n \geq 3600$.

2. Section 5.2 Problem 4.

In order to estimate f , the true fraction of smokers in a large population, Alvin selects n people at random. His estimator M_n is obtained by dividing S_n , the number of smokers in his sample, by n , i.e., $M_n = S_n/n$. Alvin chooses the sample size n to be the smallest possible number for which the Chebyshev inequality yields a guarantee that

$$\mathbf{P}(|M_n - f| \geq \epsilon) \leq \delta$$

where ϵ and δ are some prespecified tolerances. Determine how the value of n recommended by the Chebyshev inequality changes in the following cases.

- (a). The value of ϵ is reduced to half its original value.
- (b). The probability δ is reduced to half its original value.

Solution: Let $X_1, \dots, X_n$ be i.i.d. Bernoulli RVs, with success probability f and variance $\sigma^2 = f(1-f)$. The estimator M_n becomes $M_n = S_n/n = (X_1 + \dots + X_n)/n$. By Chebyshev inequality, we have

$$\mathbf{P}(|M_n - f| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2} = \frac{f(1-f)}{n\epsilon^2} \leq \frac{1}{4n\epsilon^2}$$

- (a). If ϵ is reduced to half its original value, and in order to keep the bound $1/(4n\epsilon^2)$ constant, the sample size n must be made four times larger.
 - (b). If the error probability δ is to be reduced to $\delta/2$, while keeping ϵ the same, the sample size has to be doubled.
3. From past experience, a professor knows that the test score of a student taking her final examination is a random variable with mean 75.
- (a). Give an upper bound for the probability that a student's test score will exceed 85.
 - (b). Suppose, in addition, that the professor knows that the variance of a student's test score is equal to 25. What can be said about the probability that a student will score between 65 and 85?
 - (c). How many students would have to take the examination to ensure with probability at least 0.9 that the class average would be within 5 of 75?

Solution:

- (a). With Markov inequality,

$$\mathbf{P}(X \geq 85) \leq \frac{\mathbf{E}[X]}{85} = \frac{15}{17}$$

- (b). With Chebyshev inequality,

$$\mathbf{P}(65 \leq X \leq 85) = 1 - \mathbf{P}(|X - 75| > 10) \geq 1 - \frac{25}{10^2} = \frac{3}{4}$$

(c). With Chebyshev inequality,

$$\mathbf{P}\left(\left|\frac{\sum_{i=1}^n X_i}{n} - 75\right| > 5\right) < \frac{25}{n \cdot 5^2} \leq 0.1$$

So we need $n = 10$.

4. ***Bonus.** If X is a Poisson random variable with mean λ , show that for $i < \lambda$,

$$\mathbf{P}(X \leq i) \leq \frac{e^{-\lambda}(e\lambda)^i}{i^i}$$

Hint: Use Chernoff inequality. The transform of Poisson RV is $M_X(s) = e^{\lambda(e^s - 1)}$.